

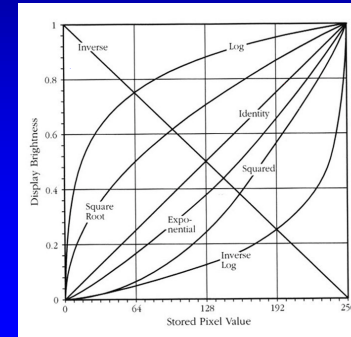
Ch. 4. Image Enhancement

Processing in the Spatial Domain

1997. 4. 15.
이영호

Contrast manipulation

- ▶ Using lookup table or LUT substitutes.
- ▶ Manipulation of the grey scale LUT
 - ➔ in terms of a transfer ft.

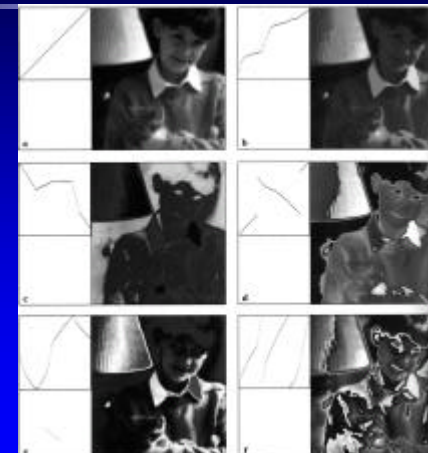


- Examples of display transfer ft. -

Image Enhancement

- ▶ Increase the visibility of portions of an image.
 - ➔ generally visibility of others are diminished.
- ▶ Image enhancement can be performed in either **spatial domain** or other domains.
- ▶ In the spatial domain, Pixel values may be modified according to rules, may be combined with or compared to others.

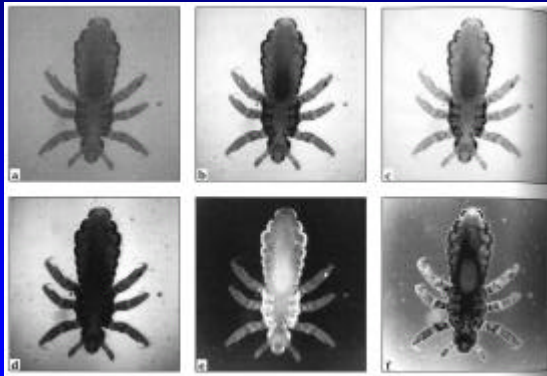
Contrast manipulation



An original image and examples of arbitrary hand-drawn transfer ft.

Contrast manipulation

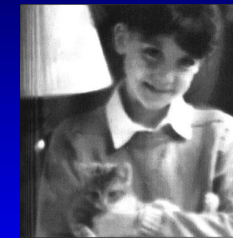
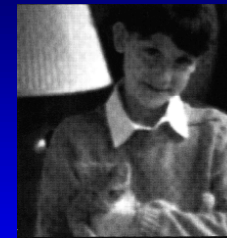
- Manipulation of the grey scale transfer function.



- a) an original image
- b) linear transfer ft.
- c) positive gamma ft.
- d) negative gamma ft.
- e) negative linear transfer ft.
- f) nonlinear transfer ft.

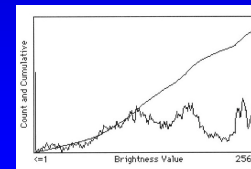
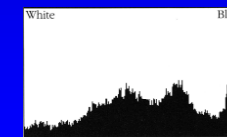
Histogram equalization

- An image with its brightness histogram.



- (a) Original image
- (b) Cumulative histogram
--> histogram equalization

$$k = \sum_{i=0}^L N_i / T$$



(a)

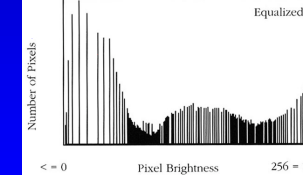
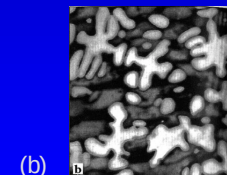
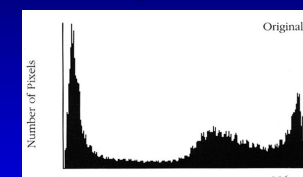
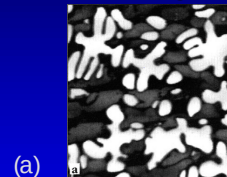
(b)

Histogram equalization

- Images have unique brightness histogram.
- Histogram equalization
 - ➔ Reassigns the brightness values of pixels based on the image histogram.
 - ➔ Individual pixels retain their brightness order, but the values are shifted, so that an equal number of pixels have each possible brightness value.

Histogram equalization

- An image with its brightness histogram.

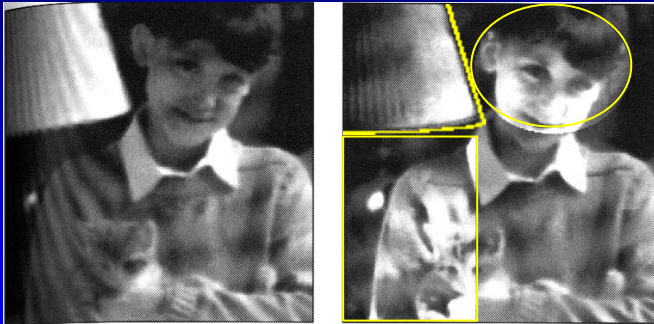


(a) Original image

(b) After histogram equalization

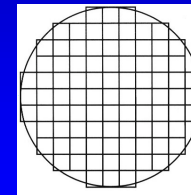
Histogram equalization

- > An image with selective equalization.

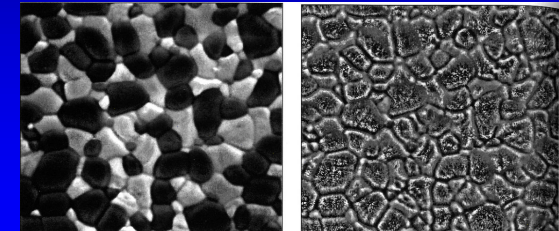


Histogram equalization

- > Local equalization using a round neighborhood.
 - Round neighborhood : Minimize directional distortion and artefacts.
 - Example of local equalization using an 11-pixel-diameter circle.
 - > Increased contrast at boundaries, and the gradients producing noise artefacts within the grains.



Round neighborhood

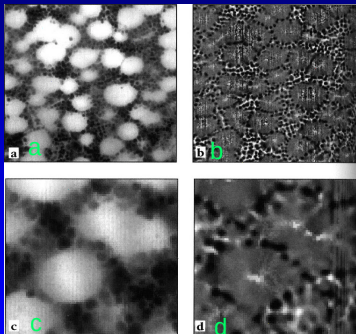


(a) Original

(b) After equalization

Histogram equalization

- > Local equalization applied to every pixel.



(a) Original image

- (b) A 9X9 neighborhood has been used.
 - this is only saved for the central pixel.
 - repeated to apply to each pixel, except those too near the edge.

(c) Enlargements to show (a)

(d) Enlargements to show (b)

Effect

장점 : Enhance edges and reveal in light and dark regions.

단점 : In the center of uniform regions it can produce artefacts and noise.

Laplacian

- > Local, or equalization neighborhood

- ➔ Making edges easier to see,
- ➔ Making the image appear sharper.

- > Laplacian (less sensitive than equalization)

$$\nabla^2 B \equiv \frac{\nabla_x^2 B}{\nabla_x^2} + \frac{\nabla_y^2 B}{\nabla_y^2}$$

- ➔ Highlights the points, and edges in the image.
- ➔ Suppress uniform and smoothly varying regions.
- ➔ A kind of good high pass filter.
- ➔ Not a good tool for demarcating edges.

Laplacian

> Laplacian operator (Ex. 3 X 3)

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

A kind of 3 X 3 smoothing **kernel**
 -- symmetrical about the center.
 -- most of them had positive weight values.

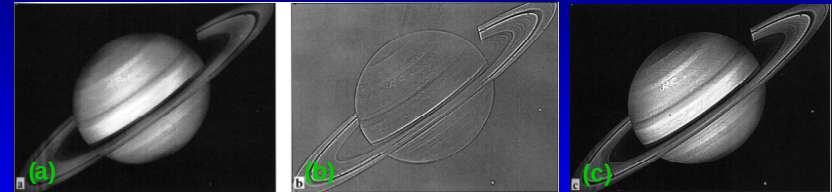
$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & +8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

A kind of 3 X 3 **Laplacian operator kernel**
 -- symmetrical about the center.
 -- does not have exclusively positive values.
 -- uniform in brightness --> grey level --> zero
 -- discontinuity area --> non-zero value

Laplacian

> Application of using a Laplacian.

--> Enhancement of the visibility of band structures in the rings.

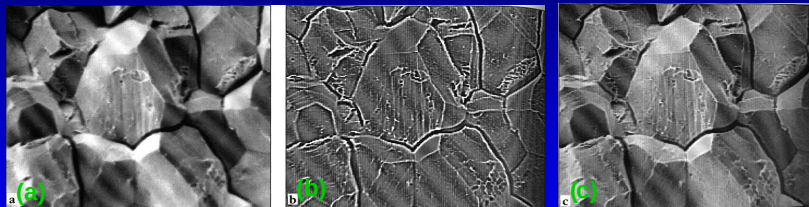


(a) Original image
 (b) Application of Laplacian operator
 (c) Subtraction of the Laplacian from the original image

Laplacian

> Application of using a Laplacian.

--> Enhancement of contrast at edges, lines and points.



- Using kernel -

$$\begin{bmatrix} +1 & +1 & +1 \\ +1 & -7 & +1 \\ +1 & +1 & +1 \end{bmatrix}$$

(a) Original image
 (b) Application of Laplacian operator
 (c) Subtraction of the Laplacian from the original image

Laplacian

> The human visual system

--> Concentrates on edges.

--> Ignores uniform regions.

> 원인

--> Presence of **Mach bands**

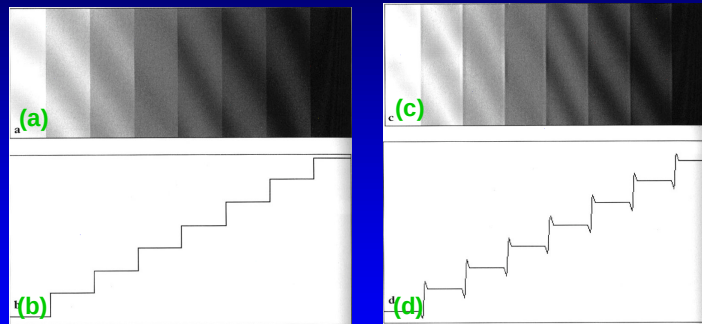
: Not perceive bands as uniform,

Sees an undershoot and overshoot on each step.

--> Presence of **local inhibition**.

Laplacian

> Illustration of *Mach bands*.



- a) Uniform grey bands b) Brightness profile of the bands
 c) Application of a **Laplacian** to (a) [This is the way eye perceives (a)]
 d) Brightness profile of image (c)

Derivatives

> **Laplacian**

--> larger response to a line than to a step and a point than a line.

> **Derivatives**

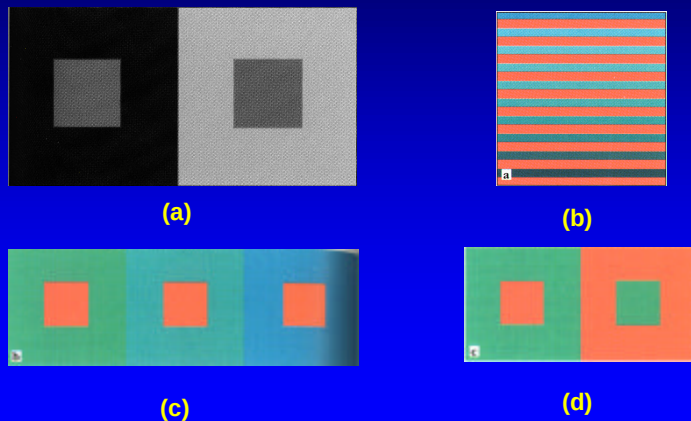
--> demarcates the boundaries.

--> enhance the visibility of small steps and other details.

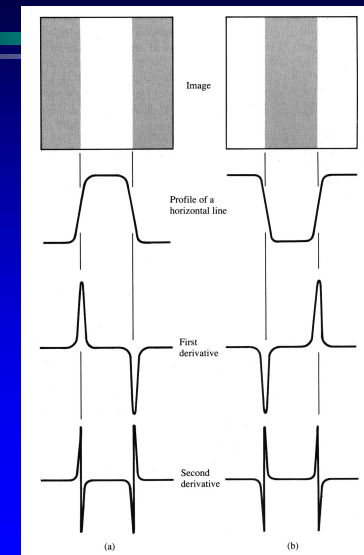
--> using 1st derivatives in two or more directions.

Laplacian

> An examples of *Local inhibition*

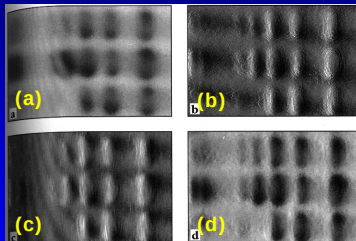


Comparison of Laplacian and Derivatives



Derivatives

> Image of a protein separation gel

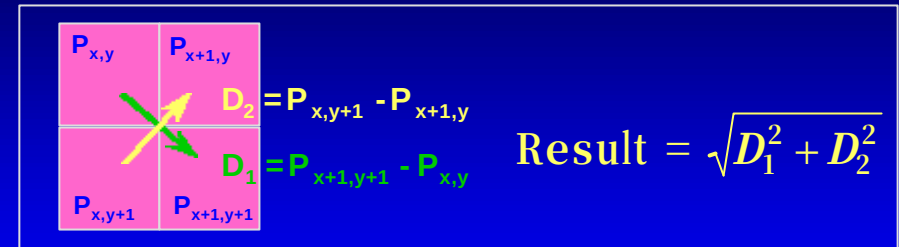


- (a) Original image
- (b) Difference between each pixel and its neighbor to the left
- (c) Vertical averaging of image (b) to reduce noise
- (d) horizontal derivatives using a 3 X 3 kernel
--> a derivative image with smoother appearance

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{bmatrix}$$

Derivatives

> Diagrams of Robert's Cross operator



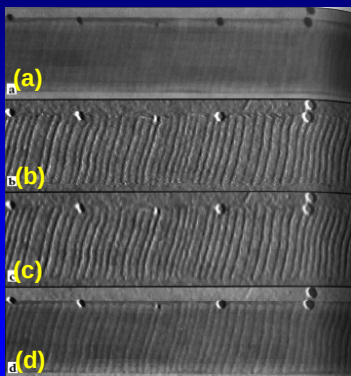
--> Two difference represent a finite approximation to the derivatives of brightness.

문제점

- Noise in the image is magnified by the single-pixel difference.
- Result is shifted by one-half of a pixel in both x and y direction.

Derivatives

> Image of tree rings in a drill core

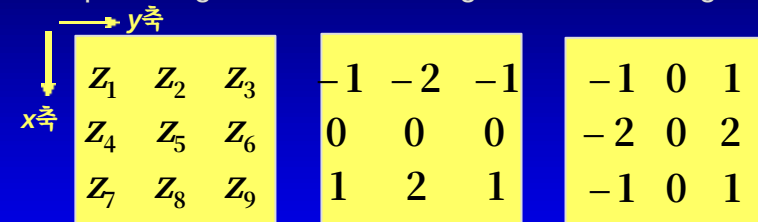


- (a) Original image
- (b) Difference between each pixel and its neighbor to the left
- (c) Vertical averaging of image (b) to reduce noise
- (d) horizontal derivatives using a 3 X 3 kernel

The Sobel and Kirsh operators

> Sobel operator

: providing both a differencing and a smoothing effect



$$G_x = (z_7 + 2z_8 + z_9) - (z_1 + 2z_2 + z_3)$$

$$G_y = (z_3 + 2z_6 + z_9) - (z_1 + 2z_4 + z_7)$$

$$\nabla f = \text{mag}(\nabla f) = [G_x^2 + G_y^2]$$

The Sobel and Kirsh operators

> Kirsh operator

: applies each of the eight orientations of the derivative kernel and keeps the **maximum value**.

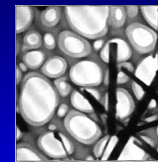
--> It requires only integer multiplication and comparison.

--> results for the magnitude of edges are similar to the **Sobel operator**.

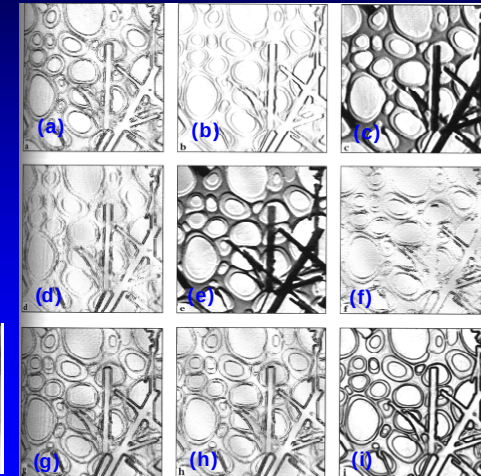
--> possible to calculate a direction value as

$$Direction = \tan^{-1}\left(\frac{\sqrt[4]{B}/\sqrt[4]{V}}{\sqrt[4]{B}/\sqrt[4]{X}}\right)$$

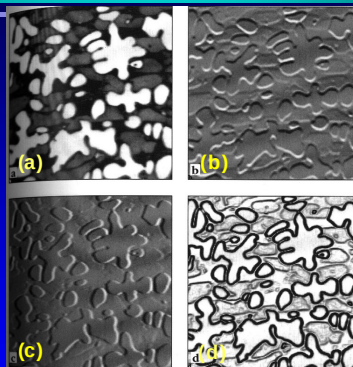
The Sobel and Kirsh operators



original image



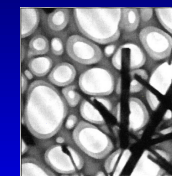
The Sobel and Kirsh operators



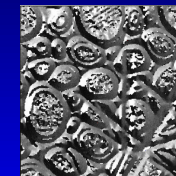
- (a) original image
- (b) image with x directional derivatives
- (c) image with y directional derivatives
- (d) Kirsh image

The Sobel and Kirsh operators

> Direction of the Sobel operator of the image



(a)



(b)

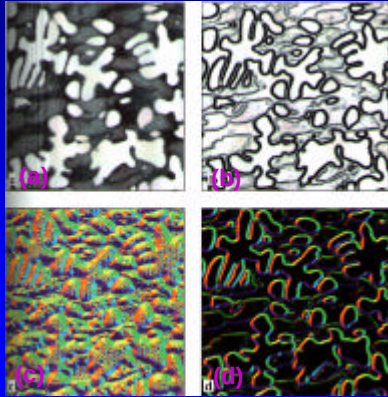
(a) original image

(b) direction of Sobel operator image

$$Direction = \tan^{-1}\left(\frac{\sqrt[4]{B}/\sqrt[4]{V}}{\sqrt[4]{B}/\sqrt[4]{X}}\right)$$

The Sobel and Kirsh operators

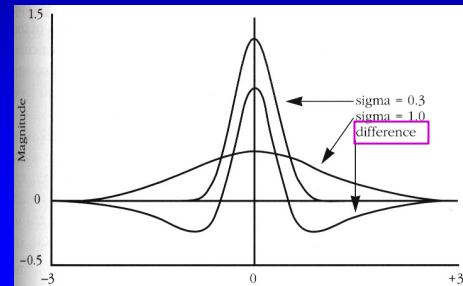
- Combining the magnitude and direction of the Sobel operator of the image



a) original grey scale image
b) Sobel gradient magnitude
c) Sobel direction (color coded)
d) product of b times c

The Sobel and Kirsh operators

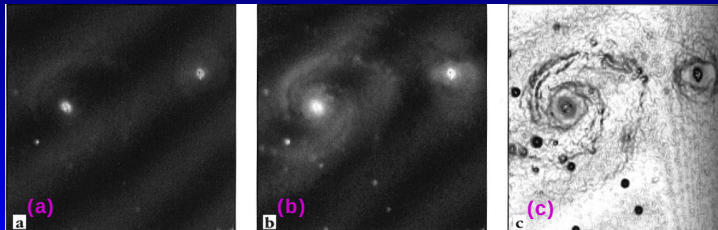
- Difference of Gaussian (D.O.G)** method
 - closely related to *unsharp masking*
 - subtraction of one smoothed version of the image from another having a different degree of smoothing



- D.O.G. operator -

The Sobel and Kirsh operators

- Enhancing an astronomical image



a) original image (brightness range too great for printing)
b) application of '*unsharp masking*' by subtracting a smoothed image
c) gradient of original image using a **Sobel operator**

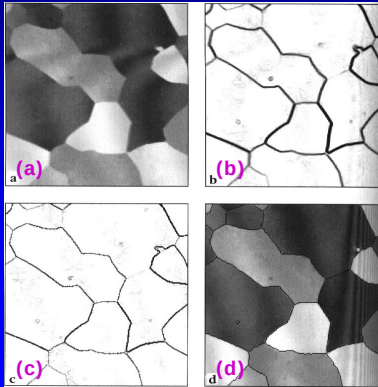
The Sobel and Kirsh operators

- Variance operator**
 - calculate sum of squares of the brightness differences for the 8 pixels surrounding each pixel in the image
 - value : very small ==> in uniform regions
large ==> a step is present

The Sobel and Kirsh operators

> Variance operator

-- Delineating boundaries between grains

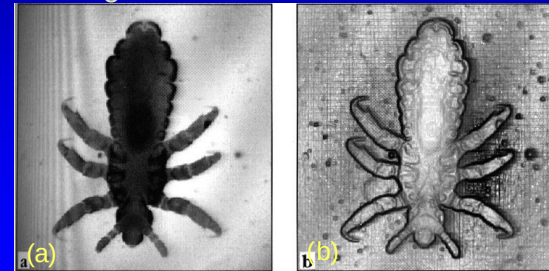


- a) original image
- b) variance edge-finding algorithm applied to image (a)
- c) gray scale skeletonization applied to image (b)
- d) thresholding and skeletonization of the boundaries to a single line of pixels produces the grain boundaries

The Sobel and Kirsh operators

> Frei and Chen algorithm

-- Advantage is sensitive to a configuration of relative pixel values independent of the magnitude of the brightness.



- (a) original image
- (b) result of Frei and Chen operator

The Sobel and Kirsh operators

> Frei and Chen algorithm

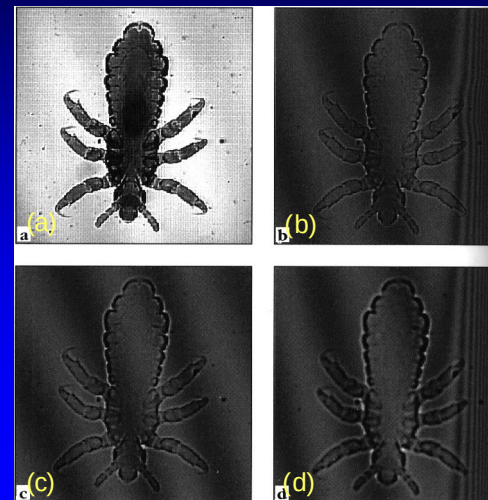
-- applies a set of *kernels* to each point in the image

-- Each *kernel* extracts one kind of behavior in the image, only a few of which are indicative of the presence of an edge.

$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 1 \\ -\sqrt{2} & 0 & \sqrt{2} \\ -1 & 0 & 1 \end{bmatrix}$	$\begin{bmatrix} \sqrt{2} & -1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & -\sqrt{2} \end{bmatrix}$	$\begin{bmatrix} -1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & -1 \end{bmatrix}$	$\begin{bmatrix} -2 & 1 & -2 \\ 1 & 4 & 1 \\ -2 & 1 & -2 \end{bmatrix}$
0	2	4	6	8
$\begin{bmatrix} -1 & -\sqrt{2} & -1 \\ 0 & 0 & 0 \\ 1 & \sqrt{2} & 1 \end{bmatrix}$	$\begin{bmatrix} 0 & -1 & \sqrt{2} \\ 1 & 0 & -1 \\ -\sqrt{2} & 1 & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$	$\begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}$	
1	3	5	7	

The Sobel and Kirsh operators

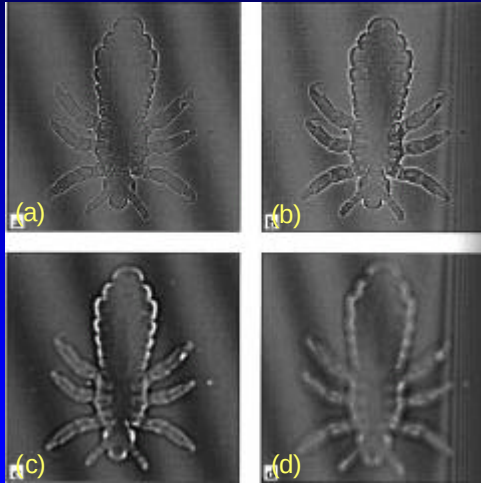
> Application of Laplacian difference operation



- a) sharpening (addition of the 3 X 3 Laplacian to the orig. image)
- b) 5 X 5 Laplacian
- c) 7 X 7 Laplacian
- d) 9 X 9 Laplacian

The Sobel and Kirsh operators

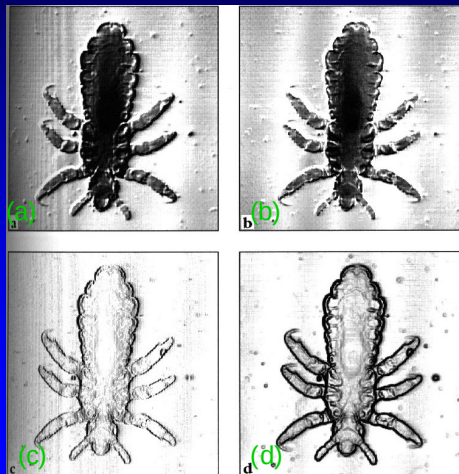
> Difference of Gaussians



- a) orig. image minus
3 X 3 Gaussian smooth
- b) difference between
 $\sigma = 0.625$ $\sigma = 1.0$
- c) difference between
 $\sigma = 1.6$ $\sigma = 1.0$
- c) difference between
 $\sigma = 4.1$ $\sigma = 2.6$

The Sobel and Kirsh operators

> Derivative edge finding



- a) single derivative
(from the upper left)
- b) brightest value at each
point after applying two
crossed derivatives
- c) Roberts' Cross operator
- d) Sobel operator